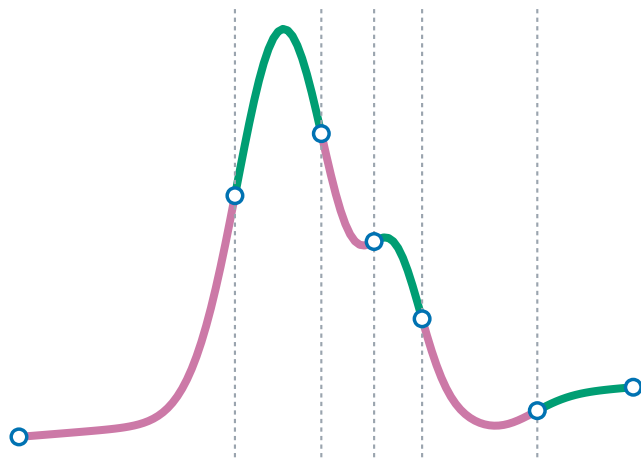
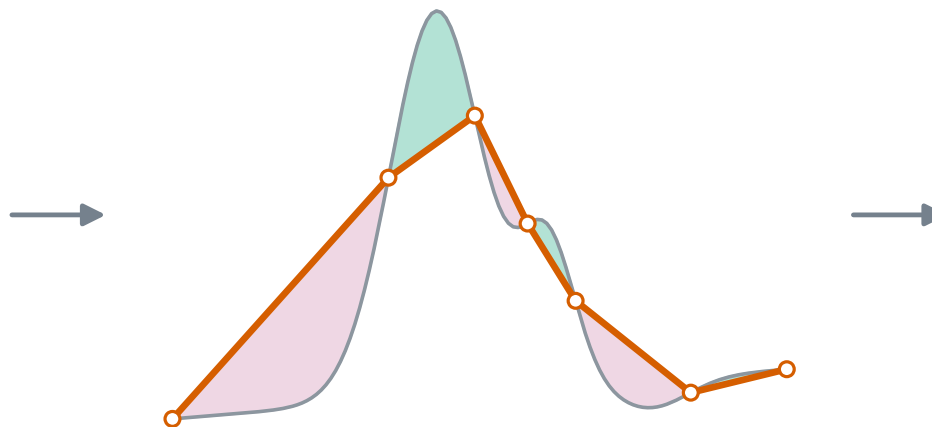


HCRD: curvature-run knots $\rightarrow$ signed chord lobes $\rightarrow$ exact hierarchy

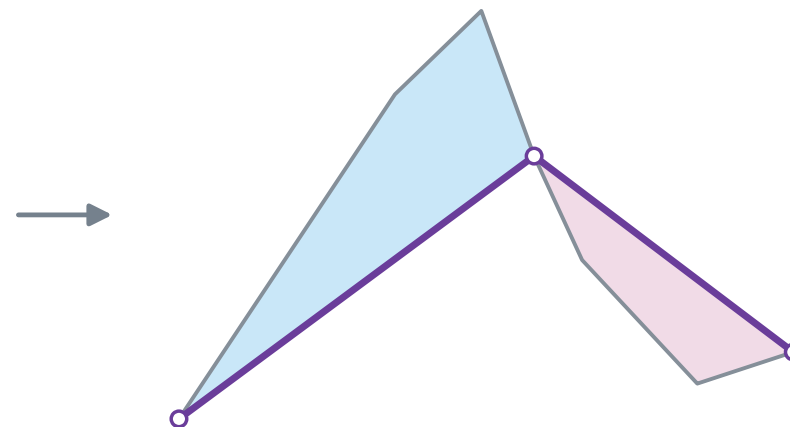
1 Find maximal curvature-sign runs



2 Join each run's endpoints



3 Recurse on retained knots



Finite • interpretable • nested • linear total knot-only work

Validated for cross-study LC-MS peak curation